ThermoFisher
SCIENTIFIC

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 27-Aug-2024 **Revision Date** 27-Aug-2024 **Revision Number** 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Neutral Buffered Formalin 10%**
Cat No. : **S60436**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

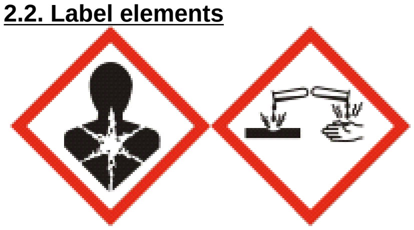
Skin Corrosion/Irritation	Category 2 (H315)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Skin Sensitization	Category 1 (H317)
Germ Cell Mutagenicity	Category 2 (H341)
Carcinogenicity	Category 1B (H350)
Specific target organ toxicity - (single exposure)	Category 2 (H371)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word **Danger**

Hazard Statements

H315 - Causes skin irritation
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H371 - May cause damage to organs
H341 - Suspected of causing genetic defects
H350 - May cause cancer

Precautionary Statements

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P280 - Wear protective gloves/protective clothing/eye protection/face protection

Additional EU labelling

Restricted to professional users

2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Water	7732-18-5	231-791-2	92.98	-
Formaldehyde	50-00-0	200-001-8	4	Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 3 (H331) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Skin Sens. 1 (H317) STOT SE 3 (H335) Muta. 2 (H341) Carc. 1B (H350)
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4		1.6	-
Methanol	67-56-1	200-659-6	1.08	Flam. Liq. 2 (H225) Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 3 (H331) STOT SE 1 (H370)
Phosphoric acid, monosodium salt	7558-80-7	EEC No. 231-449-2	0.34	-

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
Formaldehyde	Skin Corr. 1B :: C>=25% Eye Irrit. 2 :: 5%<=C<25% Skin Irrit. 2 :: 5%<=C<25% Skin Sens. 1 :: C>=0.2% STOT SE 3 :: C>=5%	-	-
Methanol	STOT Single Exp. 1 :: >= 10 STOT Single Exp. 2 :: 3 - < 10	-	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

Ingestion Clean mouth with water and drink afterwards plenty of water.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . Causes severe eye damage. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbon dioxide (CO_2), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products
Carbon oxides, Oxides of phosphorus, Sodium oxides.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before

re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Storage Class/LGK 6.1D

Switzerland - Storage of hazardous substances Storage class - SC 6.1
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
Formaldehyde	TWA: 0.37 mg/m³ (8h) TWA: 0.62 mg/m³ (8h) TWA: 0.3 ppm (8h) TWA: 0.5 ppm (8h) Skin STEL: 0.74 mg/m³ (8h) STEL: 0.6 ppm (8h)	STEL: 2 ppm 15 min STEL: 2.5 mg/m³ 15 min TWA: 2 ppm 8 hr TWA: 2.5 mg/m³ 8 hr Carc.	TWA / VME: 0.5 ppm (8 heures). for the healthcare, funeral and embalming sectors until July 11, 2024 TWA / VME: 0.3 ppm (8 heures). restrictive limit TWA / VME: 0.37 mg/m³ (8 heures). restrictive limit TWA / VME: 0.62 mg/m³ (8 heures). for the healthcare, funeral and embalming sectors until July 11, 2024 STEL / VLCT: 0.6 ppm. restrictive limit STEL / VLCT: 0.74 mg/m³. restrictive limit	STEL: 0.3 ppm 15 minuten STEL: 0.38 mg/m³ 15 minuten	STEL / VLA-EC: 0.6 ppm (15 minutos). STEL / VLA-EC: 0.74 mg/m³ (15 minutos). TWA / VLA-ED: 0.3 ppm (8 horas) TWA / VLA-ED: 0.37 mg/m³ (8 horas)
Methanol	TWA: 200 ppm 8 hr TWA: 260 mg/m³ 8 hr Skin	WEL - TWA: 200 ppm TWA; 266 mg/m³ TWA WEL - STEL: 250 ppm STEL; 333 mg/m³ STEL	TWA / VME: 200 ppm (8 heures). restrictive limit TWA / VME: 260 mg/m³ (8 heures). restrictive limit STEL / VLCT: 1000 ppm. restrictive limit: this value is not set by regulation and comes from a circular published by the Ministry of Labor. STEL / VLCT: 1300 mg/m³. restrictive limit: this value is not set by	TWA: 200 ppm 8 uren TWA: 266 mg/m³ 8 uren STEL: 250 ppm 15 minuten STEL: 333 mg/m³ 15 minuten Huid	TWA / VLA-ED: 200 ppm (8 horas) TWA / VLA-ED: 266 mg/m³ (8 horas) Piel

			regulation and comes from a circular published by the Ministry of Labor. Peau		

Component	Italy	Germany	Portugal	The Netherlands	Finland
Formaldehyde	TWA: 0.37 mg/m³ 8 ore. Time Weighted Average TWA: 0.3 ppm 8 ore. Time Weighted Average TWA: 0.62 mg/m³ 8 ore. Time Weighted Average for the health care, funeral and embalming sectors until July 11, 2024 TWA: 0.5 ppm 8 ore. Time Weighted Average for the health care, funeral and embalming sectors until July 11, 2024 STEL: 0.74 mg/m³ 15 minuti. Short-term STEL: 0.6 mg/m³ 15 minuti. Short-term Pelle	TWA: 0.3 ppm (8 Stunden). AGW - exposure factor 2 TWA: 0.37 mg/m³ (8 Stunden). AGW - exposure factor 2 TWA: 0.3 ppm (8 Stunden). MAK no irritation should occur during mixed exposure TWA: 0.37 mg/m³ (8 Stunden). MAK no irritation should occur during mixed exposure Höhepunkt: 0.6 ppm Höhepunkt: 0.74 mg/m³	STEL: 0.6 ppm 15 minutos STEL: 0.74 mg/m³ 15 minutos Ceiling: 0.3 ppm TWA: 0.3 ppm 8 horas TWA: 0.37 mg/m³ 8 horas TWA: 0.62 mg/m³ 8 horas TWA: 0.5 ppm 8 horas	STEL: 0.41 ppm 15 minuten STEL: 0.5 mg/m³ 15 minuten TWA: 0.12 ppm 8 uren TWA: 0.15 mg/m³ 8 uren	TWA: 0.3 ppm 8 tunteina TWA: 0.37 mg/m³ 8 tunteina TWA: 0.05 ppm 8 tunteina STEL: 0.6 ppm 15 minuutteina STEL: 0.74 mg/m³ 15 minuutteina
Methanol	TWA: 200 ppm 8 ore. Time Weighted Average TWA: 260 mg/m³ 8 ore. Time Weighted Average Pelle	100 ppm TWA MAK; 130 mg/m³ TWA MAKSkin absorber	STEL: 250 ppm 15 minutos TWA: 200 ppm 8 horas TWA: 260 mg/m³ 8 horas Pele	huid TWA: 100 ppm 8 uren TWA: 133 mg/m³ 8 uren	TWA: 200 ppm 8 tunteina TWA: 270 mg/m³ 8 tunteina STEL: 250 ppm 15 minuutteina STEL: 330 mg/m³ 15 minuutteina Iho

Component	Austria	Denmark	Switzerland	Poland	Norway
Formaldehyde	MAK-KZGW: 0.6 ppm 15 Minuten MAK-KZGW: 0.74 mg/m³ 15 Minuten MAK-TMW: 0.3 ppm 8 Stunden MAK-TMW: 0.37 mg/m³ 8 Stunden	TWA: 0.3 ppm 8 timer TWA: 0.37 mg/m³ 8 timer STEL: 0.74 mg/m³ 15 minutter STEL: 0.6 ppm 15 minutter	STEL: 0.6 ppm 15 Minuten STEL: 0.74 mg/m³ 15 Minuten TWA: 0.3 ppm 8 Stunden TWA: 0.37 mg/m³ 8 Stunden	STEL: 0.74 mg/m³ 15 minutach TWA: 0.37 mg/m³ 8 godzinach	TWA: 0.37 mg/m³ 8 timer TWA: 0.3 ppm 8 timer STEL: 0.74 mg/m³ 15 minutter. value from the regulation STEL: 0.6 ppm 15 minutter. value from the regulation Ceiling: 1 ppm Ceiling: 1.2 mg/m³
Methanol	Haut MAK-KZGW: 800 ppm 15 Minuten MAK-KZGW: 1040 mg/m³ 15 Minuten MAK-TMW: 200 ppm 8 Stunden MAK-TMW: 260 mg/m³ 8 Stunden	TWA: 200 ppm 8 timer TWA: 260 mg/m³ 8 timer STEL: 400 ppm 15 minutter STEL: 520 mg/m³ 15 minutter Hud	Haut/Peau STEL: 400 ppm 15 Minuten STEL: 520 mg/m³ 15 Minuten TWA: 200 ppm 8 Stunden TWA: 260 mg/m³ 8 Stunden	STEL: 300 mg/m³ 15 minutach TWA: 100 mg/m³ 8 godzinach	TWA: 100 ppm 8 timer TWA: 130 mg/m³ 8 timer STEL: 150 ppm 15 minutter. value calculated STEL: 162.5 mg/m³ 15 minutter. value calculated Hud

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Formaldehyde	TWA: 0.37 mg/m³ TWA: 0.3 ppm TWA: 0.62 mg/m³ STEL : 0.5 ppm STEL : 0.74 mg/m³ STEL : 0.6 ppm	TWA-GVI: 0.3 ppm 8 satima. except health, funeral and embalming sector TWA-GVI: 0.37 mg/m³ 8 satima. except health, funeral and embalming sector TWA-GVI: 0.5 ppm 8	TWA: 0.3 ppm 8 hr. TWA: 0.5 ppm 8 hr. for the healthcare, funeral and embalming sectors until July 11, 2024 TWA: 0.37 mg/m³ 8 hr. TWA: 0.62 mg/m³ 8 hr. for the healthcare, funeral and embalming	STEL: 0.74 mg/m³ STEL: 0.6 ppm TWA: 0.3 ppm TWA: 0.37 mg/m³	TWA: 0.37 mg/m³ 8 hodinách. except the field of health services, funeral and embalming services TWA: 0.5 mg/m³ 8 hodinách. for the field of health services, funeral and embalming services

		satima. applies to health, funeral and embalming sector applies until July 11, 2024 TWA-GVI: 0.62 mg/m^3 8 satima. applies to health, funeral and embalming sector applies until July 11, 2024 STEL-KGVI: 0.6 ppm 15 minutama. STEL-KGVI: 0.74 mg/m^3 15 minutama.	sectors until July 11, 2024 STEL: 0.6 ppm 15 min STEL: 0.738 mg/m^3 15 min STEL: 0.62 mg/m^3 15 min		Potential for cutaneous absorption Ceiling: 0.74 mg/m^3
Methanol	TWA: 200 ppm TWA: 260.0 mg/m^3 Skin notation	kože TWA-GVI: 200 ppm 8 satima. TWA-GVI: 260 mg/m^3 8 satima.	TWA: 200 ppm 8 hr. TWA: 260 mg/m^3 8 hr. STEL: 600 ppm 15 min STEL: 780 mg/m^3 15 min Skin	Skin-potential for cutaneous absorption TWA: 200 ppm TWA: 260 mg/m^3	TWA: 250 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 1000 mg/m^3

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Formaldehyde	TWA: 0.3 ppm 8 tundides. TWA: 0.37 mg/m^3 8 tundides. TWA: 0.62 mg/m^3 8 tundides. in the health, funeral and embalming sectors;valid until July 10, 2024 TWA: 0.5 ppm 8 tundides. in the health, funeral and embalming sectors;valid until July 10, 2024 STEL: 0.6 ppm 15 minutites. STEL: 0.74 mg/m^3 15 minutites.		STEL: 0.6 ppm STEL: 0.74 mg/m^3 TWA: 0.3 ppm TWA: 0.37 mg/m^3	STEL: 0.6 ppm 15 percekben. CK STEL: 0.74 mg/m^3 15 percekben. CK TWA: 0.3 ppm 8 órában. AK TWA: 0.37 mg/m^3 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 0.60 ppm STEL: 0.74 mg/m^3 TWA: 0.30 ppm 8 klukkustundum. TWA: 0.37 mg/m^3 8 klukkustundum. Skin notation
Methanol	Nahk TWA: 200 ppm 8 tundides. TWA: 250 mg/m^3 8 tundides. STEL: 250 ppm 15 minutites. STEL: 350 mg/m^3 15 minutites.	Skin notation TWA: 200 ppm 8 hr TWA: 260 mg/m^3 8 hr	skin - potential for cutaneous absorption STEL: 250 ppm STEL: 325 mg/m^3 TWA: 200 ppm TWA: 260 mg/m^3	TWA: 260 mg/m^3 8 órában. AK TWA: 200 ppm 8 órában. AK lehetséges borön keresztüli felszívódás	TWA: 200 ppm 8 klukkustundum. TWA: 260 mg/m^3 8 klukkustundum. Skin notation Ceiling: 400 ppm Ceiling: 520 mg/m^3

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Formaldehyde	STEL: 0.74 mg/m^3 STEL: 0.6 ppm TWA: 0.37 mg/m^3 TWA: 0.62 mg/m^3 TWA: 0.3 ppm TWA: 0.5 ppm	TWA: 0.3 ppm IPRD TWA: 0.37 mg/m^3 IPRD TWA: 0.62 mg/m^3 IPRD for healthcare, funeral, and embalming industries TWA: 0.5 ppm IPRD for healthcare, funeral, and embalming industries STEL: 0.74 mg/m^3 STEL: 0.6 ppm			Skin notation TWA: 0.62 mg/m^3 8 ore TWA: 0.5 ppm 8 ore STEL: 0.6 ppm 15 minute STEL: 0.74 mg/m^3 15 minute
Methanol	skin - potential for cutaneous exposure TWA: 200 ppm TWA: 260 mg/m^3	TWA: 200 ppm IPRD TWA: 260 mg/m^3 IPRD Oda	Possibility of significant uptake through the skin TWA: 200 ppm 8 Stunden TWA: 260 mg/m^3 8 Stunden	possibility of significant uptake through the skin TWA: 200 ppm TWA: 260 mg/m^3	Skin notation TWA: 200 ppm 8 ore TWA: 260 mg/m^3 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Formaldehyde	Skin notation MAC: 0.5 mg/m³	Ceiling: 0.74 mg/m³ TWA: 0.3 ppm TWA: 0.37 mg/m³	TWA: 0.62 mg/m³ 8 urah applies for health care, funeral and embalming activities until July 11, 2024 TWA: 0.5 ppm 8 urah applies for health care, funeral and embalming activities until July 11, 2024 TWA: 0.37 mg/m³ 8 urah TWA: 0.3 ppm 8 urah Koža STEL: 0.6 ppm 15 minutah STEL: 0.74 mg/m³ 15 minutah	Binding STEL: 0.6 ppm 15 minuter Binding STEL: 0.74 mg/m³ 15 minuter TLV: 0.3 ppm 8 timmar. NGV TLV: 0.37 mg/m³ 8 timmar. NGV Hud	
Methanol	TWA: 5 mg/m³ 1250 Skin notation MAC: 15 mg/m³	Potential for cutaneous absorption TWA: 200 ppm TWA: 260 mg/m³	TWA: 200 ppm 8 urah TWA: 260 mg/m³ 8 urah Koža STEL: 800 ppm 15 minutah STEL: 1040 mg/m³ 15 minutah	Indicative STEL: 250 ppm 15 minuter Indicative STEL: 350 mg/m³ 15 minuter TLV: 200 ppm 8 timmar. NGV TLV: 250 mg/m³ 8 timmar. NGV Hud	Deri TWA: 200 ppm 8 saat TWA: 260 mg/m³ 8 saat
Phosphoric acid, monosodium salt	MAC: 10 mg/m³				

Biological limit values
List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Methanol			Methanol: urine end of shift	Methanol: 15 mg/L urine end of shift	Methanol: 15 mg/L urine (end of shift) Methanol: 15 mg/L urine (for long-term exposures: at the end of the shift after several shifts)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Methanol					Methanol: 6 mg/L urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Methanol			Methanol: 30 mg/L urine end of exposure or work shift Methanol: 30 mg/L urine after all work shifts for long-term exposure		

Monitoring methods
MDHS78 Formaldehyde in air. Laboratory method using a diffusive sampler, solvent desorption and high performance liquid chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)
See table for values

Component	Acute effects local	Acute effects	Chronic effects local	Chronic effects

	(Dermal)	systemic (Dermal)	(Dermal)	systemic (Dermal)
Formaldehyde 50-00-0 (4)			DNEL = 37µg/cm2	DNEL = 240mg/kg bw/day
Methanol 67-56-1 (1.08)		DNEL = 20mg/kg bw/day		DNEL = 20mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Formaldehyde 50-00-0 (4)	DNEL = 0.75mg/m³		DNEL = 0.375mg/m³	DNEL = 9mg/m³
Methanol 67-56-1 (1.08)	DNEL = 130mg/m³	DNEL = 130mg/m³	DNEL = 130mg/m³	DNEL = 130mg/m³

Predicted No Effect Concentration (PNEC)
See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Formaldehyde 50-00-0 (4)	PNEC = 0.44mg/L	PNEC = 2.3mg/kg sediment dw	PNEC = 4.44mg/L	PNEC = 0.19mg/L	PNEC = 0.2mg/kg soil dw
Methanol 67-56-1 (1.08)	PNEC = 20.8mg/L	PNEC = 77mg/kg sediment dw	PNEC = 1540mg/L	PNEC = 100mg/L	PNEC = 100mg/kg soil dw
Phosphoric acid, monosodium salt 7558-80-7 (0.34)	PNEC = 0.05mg/L		PNEC = 0.5mg/L	PNEC = 50mg/L	

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Formaldehyde 50-00-0 (4)	PNEC = 0.44mg/L	PNEC = 2.3mg/kg sediment dw			
Methanol 67-56-1 (1.08)	PNEC = 2.08mg/L	PNEC = 7.7mg/kg sediment dw			
Phosphoric acid, monosodium salt 7558-80-7 (0.34)	PNEC = 0.005mg/L				

8.2. Exposure controls

Engineering Measures
Ensure that eyewash stations and safety showers are close to the workstation location.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection Goggles (European standard - EN 166)

Hand Protection Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber Nitrile rubber Neoprene PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** Particulates filter conforming to EN 143
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Particle filtering: EN149:2001 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance		
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	100 °C / 212 °F	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	**Method -** No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
Formaldehyde	-0.35	
Methanol	-0.74	
Vapor Pressure	No data available	
Density / Specific Gravity	1.017	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	Not applicable (liquid)	

9.2. Other information

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Carbon oxides. Oxides of phosphorus. Sodium oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Formaldehyde	500 mg/kg (Rat)	LD50 = 270 mg/kg (Rabbit)	0.578 mg/L (Rat) 4 h
Methanol	LD50 = 1187 – 2769 mg/kg (Rat)	LD50 = 17100 mg/kg (Rabbit)	LC50 = 128.2 mg/L (Rat) 4 h
Phosphoric acid, monosodium salt	LD50 = 8290 mg/kg (Rat)	LD50 > 7940 mg/kg (Rabbit)	LC50 > 0.83 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available
Skin Category 1

Component	Test method	Test species	Study result
Formaldehyde 50-00-0 (4)	Skin sensitization Test method Patch Test Respiratory sensitization in vitro	Man guinea pig	Sensitizer Sensitization
Methanol 67-56-1 (1.08)	OECD Test Guideline 406 Guinea Pig Maximisation Test	guinea pig	non-sensitising

	(GPMT)		

May cause sensitization by skin contact

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Formaldehyde	Carc Cat. 1B	Cat 3		Group 1

(g) reproductive toxicity; No data available

Component	Test method	Test species / Duration	Study result
Methanol 67-56-1 (1.08)	OECD Test Guideline 416	Rat / Inhalation 2 Generation	NOAEC = 1.3 mg/l (air)

(h) STOT-single exposure; Category 2

Results / Target organs Optic nerve, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects .

Component	Freshwater Fish	Water Flea	Freshwater Algae
Formaldehyde	Leuciscus idus: LC50 = 15 mg/L 96h	EC50 = 20 mg/L 96h EC50 = 2 mg/L 48h	EC50 (72h) = 4.89 mg/L (Desmodesmus subspicatus)
Methanol	Pimephales promelas: LC50 > 10000 mg/L 96h	EC50 > 10000 mg/L 24h	

Component	Microtox	M-Factor
Methanol	EC50 = 39000 mg/L 25 min EC50 = 40000 mg/L 15 min EC50 = 43000 mg/L 5 min	

12.2. Persistence and degradability

No information available

Component	Degradability
Formaldehyde 50-00-0 (4)	Readily biodegradable (OECD guideline 301A, 301C and 301D) under aerobic and anaerobic conditions.

Methanol 67-56-1 (1.08)	DT50 ~ 17.2d >94% after 20d

12.3. Bioaccumulative potential No information available

Component	log Pow	Bioconcentration factor (BCF)
Formaldehyde	-0.35	No data available
Methanol	-0.74	<10 dimensionless

12.4. Mobility in soil No information available

12.5. Results of PBT and vPvB assessment No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

ADR Not regulated

14.1. UN number
14.2. UN proper shipping name

14.3. Transport hazard class(es)
14.4. Packing group

IATA Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-
Formaldehyde	50-00-0	200-001-8	-	-	X	X	KE-17074	X	X
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	-	-	-	X	X	-	X	-
Methanol	67-56-1	200-659-6	-	-	X	X	KE-23193	X	X
Phosphoric acid, monosodium salt	7558-80-7	231-449-2	-	-	X	X	KE-31577	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Water	7732-18-5	X	ACTIVE	X	-	X	X	X
Formaldehyde	50-00-0	X	ACTIVE	X	-	X	X	X
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	-	-	-	-	X	X	X
Methanol	67-56-1	X	ACTIVE	X	-	X	X	X
Phosphoric acid, monosodium salt	7558-80-7	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Formaldehyde	50-00-0	-	Use restricted. See entry 72. (see link for restriction details) Use restricted. See entry 77.	-

			(see link for restriction details) Use restricted. See entry 28. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	-	-	-
Methanol	67-56-1	-	Use restricted. See entry 69. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	-
Phosphoric acid, monosodium salt	7558-80-7	-	-	-

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Water	7732-18-5	Not applicable	Not applicable
Formaldehyde	50-00-0	5 tonne	50 tonne
Phosphoric acid, disodium salt, dodecahydrate	10039-32-4	Not applicable	Not applicable
Methanol	67-56-1	500 tonne	5000 tonne
Phosphoric acid, monosodium salt	7558-80-7	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 3 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Formaldehyde	WGK 3	Krebserzeugende Stoffe - : 5 mg/m³ (Massenkonzentration)
Phosphoric acid, disodium salt, dodecahydrate	WGK1	
Methanol	WGK 2	Class I : 20 mg/m³ (Massenkonzentration)
Phosphoric acid, monosodium	WGK1	

salt		

Component	France - INRS (Tables of occupational diseases)
Formaldehyde	Tableaux des maladies professionnelles (TMP) - RG 43
Methanol	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Formaldehyde 50-00-0 (4)		Group I	
Methanol 67-56-1 (1.08)	Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H315 - Causes skin irritation
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H371 - May cause damage to organs
H341 - Suspected of causing genetic defects
H350 - May cause cancer
H225 - Highly flammable liquid and vapor
H301 - Toxic if swallowed
H311 - Toxic in contact with skin
H314 - Causes severe skin burns and eye damage
H331 - Toxic if inhaled
H335 - May cause respiratory irritation
H370 - Causes damage to organs

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
MARPOL - International Convention for the Prevention of Pollution from Ships
OECD - Organisation for Economic Co-operation and Development
ATE - Acute Toxicity Estimate
BCF - Bioconcentration factor
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

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This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer
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End of Safety Data Sheet